Mean, Variance and Bias

Machine Learning Basics

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Learning Objectives

By the end of this topic, students will be able to:

- Understand the meaning of mean.
- Calculate variance for a given dataset.
- Understand bias in machine learning models.
- Differentiate low bias, high bias, low variance and high variance.
- Relate bias and variance with underfitting and overfitting.

Mean

Mean is also called the average.
It gives the central value of a set of numbers.

$$\bar{x} = \frac{1}{N} \sum_{i=1}^{N} x_i$$

where

- $\bar{x}$ is the mean,
- x_i is the i^{th} observation,
- N is the total number of observations.

Example: Mean

Suppose a student gets the following five test scores:

$$70,\ 72,\ 68,\ 74,\ 66$$

$$\bar{x} = \frac{70 + 72 + 68 + 74 + 66}{5}$$

$$\bar{x} = \frac{350}{5} = 70$$

$$\boxed{\text{Mean} = 70}$$

Variance

Variance tells us how spread out the values are from the mean.

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$$

- Low variance: values are close to the mean.
- High variance: values are far from the mean.
- If all values are the same, variance is zero.

Example: Variance

Scores:

$$70,\ 72,\ 68,\ 74,\ 66$$

Mean:

$$\bar{x} = 70$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
70	0	0
72	2	4
68	-2	4
74	4	16
66	-4	16

$$\sigma^2 = \frac{0 + 4 + 4 + 16 + 16}{5} = 8$$

Bias in Machine Learning

Bias tells us how far the average model prediction is from the true value.

$$\text{Bias}(x) = \mathbb{E}[\hat{f}(x)] - f(x)$$

where

- $f(x)$ is the true function,
- $\hat{f}(x)$ is the model prediction,
- $\mathbb{E}[\hat{f}(x)]$ is the average prediction.

$\boxed{\text{High bias means the model makes systematic errors.}}$

Variance of a Model

Model variance tells us how much predictions change when the training data changes.

$$\mathrm{Var}(\hat{f}(x)) = \mathbb{E}\left[\left(\hat{f}(x) - \mathbb{E}[\hat{f}(x)]\right)^2\right]$$

- Low variance: model gives stable predictions.
- High variance: model predictions change a lot.

Problem Context

Assume the true function is

$$f(x) = 5x + 50$$

where

- x is the number of hours studied,
- $f(x)$ is the exam score.

If a student studies for 4 hours,

$$f(4) = 5(4) + 50 = 70$$

$$\boxed{\text{True score} = 70}$$

Case 1: Low Bias and Low Variance

Model predictions:

$$69.9, \quad 69.7, \quad 70.4$$

Mean prediction:

$$\mathbb{E}[\hat{f}(4)] = \frac{69.9 + 69.7 + 70.4}{3} = 70$$

Bias:

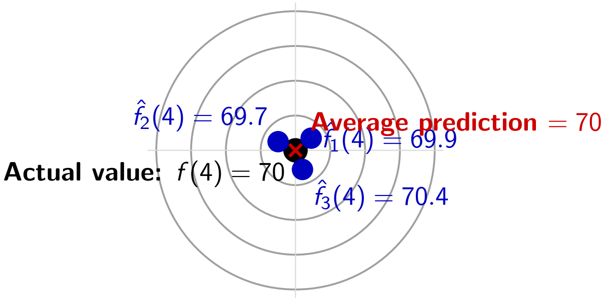
$$\text{Bias} = 70 - 70 = 0$$

Variance:

$$\text{Var} = \frac{(69.9 - 70)^2 + (69.7 - 70)^2 + (70.4 - 70)^2}{3}$$

$$\text{Var} \approx 0.087$$

Case 1: Low Bias and Low Variance



Low Bias and Low Variance

Predictions are close to the actual value and close to one another

Case 2: High Bias and Low Variance

Model predictions:

$$65, \quad 65, \quad 65$$

Mean prediction:

$$\mathbb{E}[\hat{f}(4)] = 65$$

Bias:

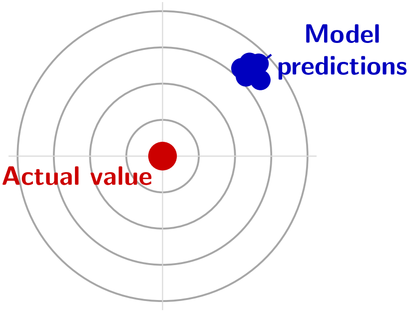
$$\text{Bias} = 65 - 70 = -5$$

$$\text{Bias}^2 = 25$$

Variance:

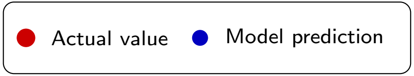
$$\text{Var} = 0$$

Case 2: High Bias and Low Variance



High Bias and Low Variance

Predictions are close to one another but far from the actual value



Case 3: Low Bias and High Variance

Model predictions:

$$67, \quad 73, \quad 71$$

Mean prediction:

$$\mathbb{E}[\hat{f}(4)] = \frac{67 + 73 + 71}{3} = 70.33$$

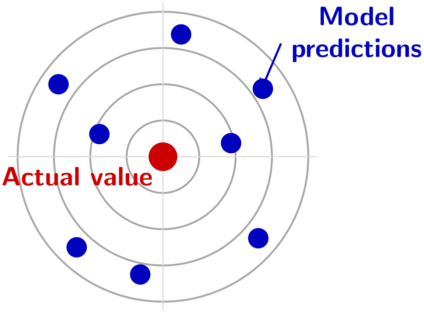
Bias:

$$\text{Bias} = 70.33 - 70 = 0.33$$

Variance:

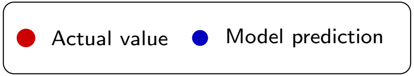
$$\text{Var} \approx 6.22$$

$\boxed{\text{Average prediction is good, but predictions vary a lot}}$



Low Bias and High Variance

Predictions are centred around the actual value but widely spread



Case 4: High Bias and High Variance

Model predictions:

$$60, \quad 65, \quad 75$$

Mean prediction:

$$\mathbb{E}[\hat{f}(4)] = \frac{60 + 65 + 75}{3} = 66.67$$

Bias:

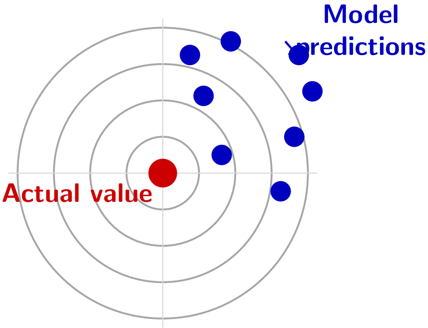
$$\text{Bias} = 66.67 - 70 = -3.33$$

$$\text{Bias}^2 \approx 11.11$$

Variance:

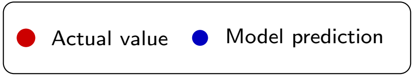
$$\text{Var} \approx 38.89$$

Case 4: High Bias and High Variance



High Bias and High Variance

Predictions are far from the actual value and widely spread



Bias and Variance Summary

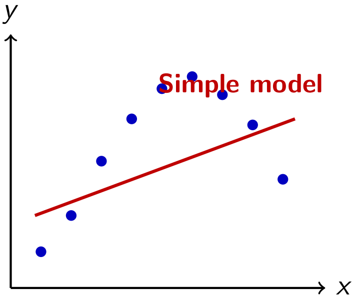
Bias	Variance	Meaning
Low	Low	Accurate and stable
High	Low	Underfitting
Low	High	Overfitting (Sensitive to data changes)
High	High	Wrong and unstable

Underfitting

Underfitting happens when the model is too simple.

$$\boxed{\text{Underfitting} \Rightarrow \text{High Bias}}$$

- It fails to learn the true pattern.
- Training and testing accuracy is low.
- Bias is high.



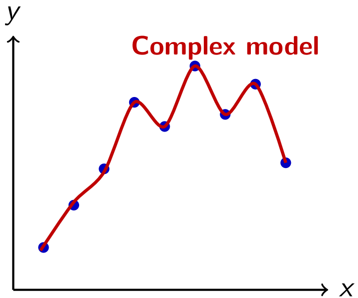
The model fails to learn

Overfitting

Overfitting happens when the model learns the training data too closely.

$$\boxed{\text{Overfitting} \Rightarrow \text{High Variance}}$$

- It learns noise also.
- Training accuracy is very high and testing accuracy is low.
- Variance is high.



The model closely follows every training observation

Training, Validation and Testing

Training Data

- Used to train the model.
- Model parameters are learned.
- Examples: weights and biases.

Validation Data

- Used during model development.
- Hyperparameters are tuned.
- Helps in model selection.

Test Data

- Used after model training.
- Measures generalization.
- Must remain unseen.

$\boxed{\text{Good generalization} \Rightarrow \text{Good performance on unseen test data}}$

Model Generalization

Model condition	Training performance	Testing performance	Bias	Variance
Good generalization	Good	Good	Balanced	Balanced
Underfitting	Poor	Poor	High	Usually low
Overfitting	Very good	Poor	Low	High

Objective

The objective is not merely to obtain high training accuracy. The model must also perform well on unseen test data.

Bias-Variance Decomposition

Goal: Measure the prediction error of a machine learning model.

Suppose the true relationship between the input and output is

$$y = f(x) + \varepsilon$$

where

- $f(x)$: True function
- ε : Random factors or noise

  Let us assume
- $E[\varepsilon] = 0$ (since noise is independent of output)
- $Var(\varepsilon) = \sigma_\varepsilon^2$

Our objective is to learn an approximation $\hat{f}(x)$ that is as close as possible to the true function $f(x)$.

Mean Squared Error (MSE)

The prediction error is measured using the Mean Squared Error (MSE).

$$MSE(x) = E[(y - \hat{f}(x))^2]$$

Substitute

$$y = f(x) + \varepsilon$$

into the MSE expression,

$$MSE(x) = E[(f(x) + \varepsilon - \hat{f}(x))^2].$$

Using

$$(a + b)^2 = a^2 + b^2 + 2ab,$$

we obtain

$$MSE(x) = E[(f(x) - \hat{f}(x))^2] + \quad E[\varepsilon^2] + 2E[(f(x) - \hat{f}(x))\varepsilon].$$

Simplifying the MSE

The cross-product term is

$$2E[(f(x) - \hat{f}(x))\varepsilon].$$

Since the noise is random, $E[\varepsilon] = 0$.
Therefore,

$$2E[(f(x) - \hat{f}(x))\varepsilon] = 0.$$

Also,

$$E[\varepsilon^2] = \sigma_\varepsilon^2.$$

Hence,

$$\boxed{MSE(x) = E[(f(x) - \hat{f}(x))^2] + \sigma_\varepsilon^2}$$

The remaining model error is now decomposed into Bias and Variance.

Separating the Prediction Error

After removing the noise term, we have

$$E\left[(f(x) - \hat{f}(x))^2\right].$$

Here,

- $f(x)$: True function
- $\hat{f}(x)$: Prediction made by the model

This expression measures the error between the true function and the predicted function.

Question: Why does this prediction error occur?

- The average prediction may be different from the true function (**Bias**).
- The prediction may vary for different training datasets (**Variance**).

To separate these two sources of error, we introduce the average prediction $E[\hat{f}(x)]$.

Adding and Subtracting the Average Prediction

We add and subtract the same quantity $E[\hat{f}(x)]$.

$$f(x) - \hat{f}(x) = f(x) - E[\hat{f}(x)] + E[\hat{f}(x)] - \hat{f}(x)$$

Notice that

$$+E[\hat{f}(x)] - E[\hat{f}(x)] = 0.$$

Therefore, the expression remains unchanged.

Meaning of the Two Terms

After adding and subtracting the average prediction,

$$f(x) - \hat{f}(x) = \underbrace{f(x) - E[\hat{f}(x)]}_{\text{Bias}} + \underbrace{E[\hat{f}(x)] - \hat{f}(x)}_{\text{Variance}}.$$

Bias

- Difference between the true function and the average prediction.
- Indicates how accurately the model learns the true relationship.

Variance

- Difference between the model prediction and the average prediction.
- Indicates how much the model changes for different training datasets.

Bias-Variance Decomposition

Starting from the model error,

$$E\left[\left(f(x) - \hat{f}(x)\right)^2\right]$$

Add and subtract the average prediction $E[\hat{f}(x)]$,

$$= E\left[\left(f(x) - E[\hat{f}(x)] + E[\hat{f}(x)] - \hat{f}(x)\right)^2\right]$$

we obtain

$$E\left[\left(f(x) - \hat{f}(x)\right)^2\right] = E\left[\left(f(x) - E[\hat{f}(x)]\right)^2\right]$$

$$+ E\left[\left(E[\hat{f}(x)] - \hat{f}(x)\right)^2\right]$$

Removing the Cross Product

Consider the cross-product term,

$$2E\left[\left(f(x) - E[\hat{f}(x)]\right)\left(E[\hat{f}(x)] - \hat{f}(x)\right)\right].$$

since $[f(x) - E[\hat{f}(x)]$ is a constant, it can be taken outside the expectation,

$$= 2\left(f(x) - E[\hat{f}(x)]\right) E\left[E[\hat{f}(x)] - \hat{f}(x)\right].$$

Now,

$$E\left[E[\hat{f}(x)] - \hat{f}(x)\right] = E[\hat{f}(x)] - E[\hat{f}(x)] = 0.$$

Hence,

$$\boxed{2E\left[\left(f(x) - E[\hat{f}(x)]\right)\left(E[\hat{f}(x)] - \hat{f}(x)\right)\right] = 0.}$$

Bias and Variance

Therefore,

$$E\left[\left(f(x)-\hat{f}(x)\right)^2\right] = \left(f(x)-E[\hat{f}(x)]\right)^2 + E\left[\left(\hat{f}(x)-E[\hat{f}(x)]\right)^2\right].$$

where

$$\boxed{\text{Bias} = f(x)-E[\hat{f}(x)]}$$

and

$$\boxed{\text{Variance} = E\left[\left(\hat{f}(x)-E[\hat{f}(x)]\right)^2\right].}$$

Final Bias-Variance Decomposition

Adding the irreducible noise,

$$E[\varepsilon^2] = \sigma_\varepsilon^2,$$

the Mean Squared Error becomes

$$\boxed{MSE(x) = \underbrace{\left(f(x) - E[\hat{f}(x)]\right)^2}_{\text{Bias}^2} + \underbrace{E\left[\left(\hat{f}(x) - E[\hat{f}(x)]\right)^2\right]}_{\text{Variance}} + \underbrace{\sigma_\varepsilon^2}_{\text{Noise}}}$$

Interpretation

- Bias: Error due to an overly simple model (underfitting).
- Variance: Error due to sensitivity to different training datasets (overfitting).
- Noise: Inherent randomness in the data that cannot be eliminated.

Thank You